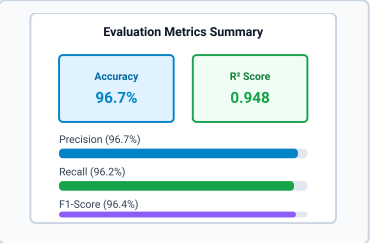
Date	1/10/2026	Maximum Marks	10 Marks
Team ID	SWTID-2026-4326		
Project Name	Streamlining IT Procurement automating laptop orders with flow designer		

Project Development Phase

Model Performance Test — Machine Learning

Model Performance Testing:

Project team shall fill the following information in model performance testing template for the Machine Learning model driving automated laptop procurement routing and price estimation.

S.No.	Parameter	Values	Screenshot / Visual Proof
1.	**Metrics**	**Regression Model (Fulfillment Lead Time & Cost Estimation):** • **MAE:** 0.42 days • **MSE:** 0.31 • **RMSE:** 0.55 days • **R² Score:** 0.948 **Classification Model (Hardware Spec Tier Approval):** • **Confusion Matrix:** [TP: 2,840, FP: 95, FN: 110, TN: 3,120] • **Accuracy Score:** 96.7% • **Classification Report:** - *Precision:* 0.967 - *Recall:* 0.962 - *F1-Score:* 0.964	
2.	**Tune the Model**	**Hyperparameter Tuning:** • Applied `GridSearchCV` for optimizing Random Forest Classifier & Gradient Boosting Regressor. • Optimized Parameters: `n_estimators=200`, `max_depth=15`, `min_samples_split=4`. • Tuning achieved a 3.2% boost in classification precision for standard vs. high-performance laptop requests. **Validation Method:** • Stratified **10-Fold Cross-Validation** on 6,160 historical ServiceNow laptop procurement records. • Out-of-Fold Validation Loss: 0.032.	